INTRODUCING SHELL RIMULA HEAVY DUTY DIESEL ENGINE OILS

Shell Rimula
ENERGISED PROTECTION

Project and Job Number 201243_1004997_Brochure_VerA.indd				1	2
Publication A4 8 Page Brochure	**Size** 297x210				
Operator Bill White	**Date** 01/09/10			3	4
Notes Shell Rimula	**Version** Proof 03				

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WORLD-CLASS PRODUCT TECHNOLOGY – PROVIDES ENERGISED PROTECTION

Technology That Adapts

From reliable products using well-proven formulation chemistry to unique and exclusive technologies for the latest low-emissions, severe-duty engines, we can offer the right performance and protection.
Built around a deep knowledge of the operation and lubrication needs of heavy duty engines, our scientists and engineers have developed products built on technology that adapts chemically and physically to the changing conditions within the engine, ensuring your engines keep running, however you choose to run your equipment, or your business.

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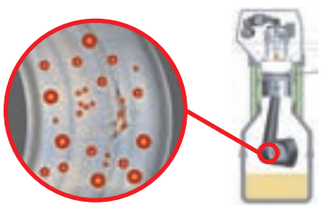
ABOUT ENERGISED PROTECTION

Shell Rimula oils are ready for action when you need them: Shell Rimula technology lets you choose the level of protection you need in three critical areas — the areas that threaten engines most.

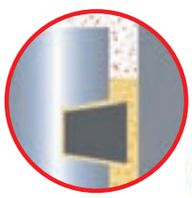
1. ACID CONTROL

Acids formed when diesel fuel is burned can escape into your oil, where they corrode parts of your engine, such as the bearings pictured here.

Acid attack on bearings can cause engine failure.



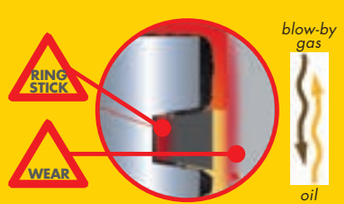
Shell Rimula oils contain powerful neutralising agents that protect the engine from attack.



Oil neutralises exhaust-gas acids as they flow past the piston and rings.

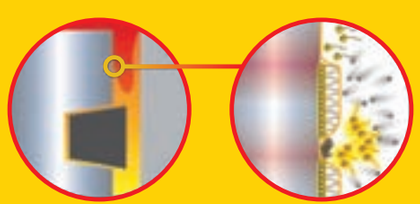
2. DEPOSIT CONTROL

Buildup of excessive piston deposits can result in ring sticking and abrasion, and cylinder wear. This shortens engine life and increases oil consumption.



Shell Rimula Energised Protection oils contain active molecules that block or carry away particles that can cause deposits.

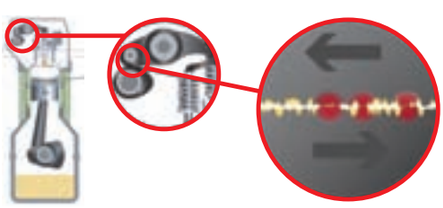
Active molecules repel or capture deposits forming carbon.



3. WEAR CONTROL

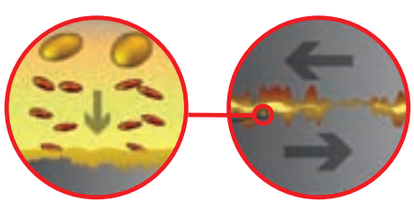
Keeping moving metal surfaces separated is a vital function of an oil. Failure can result in "hot spots" and wear.

Hot spots when lubrication breaks down.



Shell Rimula oils contain molecules that transform into protective films that keep metal surfaces apart, preventing wear.

Protective films formed by anti-wear molecules in high-stress areas.



YOUR PROTECTION - OUR EXPERTISE

Our technologists understand how to design specific protection levels into an oil to meet specific engine and operating conditions. The Shell Rimula Energised Protection range of oils are the culmination of their expertise and skill – offering a full range of protection and performance possibilities to operators of transport equipment. Let's examine these products in detail. Shell Rimula Energised Protection oils contain active molecules that block or carry away particles that can cause deposits.

THE SHELL RIMULA RANGE

Our product selection is easy to understand and use. Our portfolio of Energised Protectior vehicles fitted with diesel particulate filters (DPF) and catalysts.

PROTECTION LEVEL	QUALITY MONOGRADE OILS	ADVANCED MULTIGI
6		Shell Rimula R6 M, ME ▪ Maintenance Saving ▪ Energy Saving
5		BETTER DEPOSI Shell Rimula R5 E ▪ Energy Saving
4		BETTER ACID & WE Shell Rimula R4 ▪ Unique Active Technology ▪ Modern
3		BETTER DEPOSI Shell Rimula R3 X Triple Action – Resists: ▪ Acids ▪ Deposi
2	Shell Rimula R2 ▪ Tough, long-lasting action	BETTER ACID & WE BETTER DEPOSIT CONTROL Shell Rimula R2 Extra ▪ Turbocharger Performance

WHATEVER YOUR NEEDS, THERE'S A SHELL RIMULA OIL TO MEET IT

TRANSPORT AND LOGISTICS
Our family of oils has a solution that is right for transport and logistics operators, from cost-effective routine maintenance products such as Shell Rimula R3 X, through products that help minimise operating costs by extending maintenance or fuel-economy performance, such as Shell Rimula R6 M and ME.

And you'll be ready for tomorrow with our latest low-emissions compatible engine oils that help keep the latest Euro IV and V trucks in peak condition, such as Shell Rimula R6 LME.

BUS AND COACH
As one of the leading suppliers of engine oil technology to bus and coach operators in Europe, we can help identify the right product to meet your operating needs. For example, Shell Rimula R6 LM is an ideal solution for operators of mixed diesel and CNG powered vehicles — one oil for the whole fleet. It protects your exhaust particulate filters as well.

TRACTOR, A
Shell's exten
engine oils t
agriculture a
can be sure
round prote

With a wide
R3 X are the
on- and off-l
leading OEM
our new pro
the next ger

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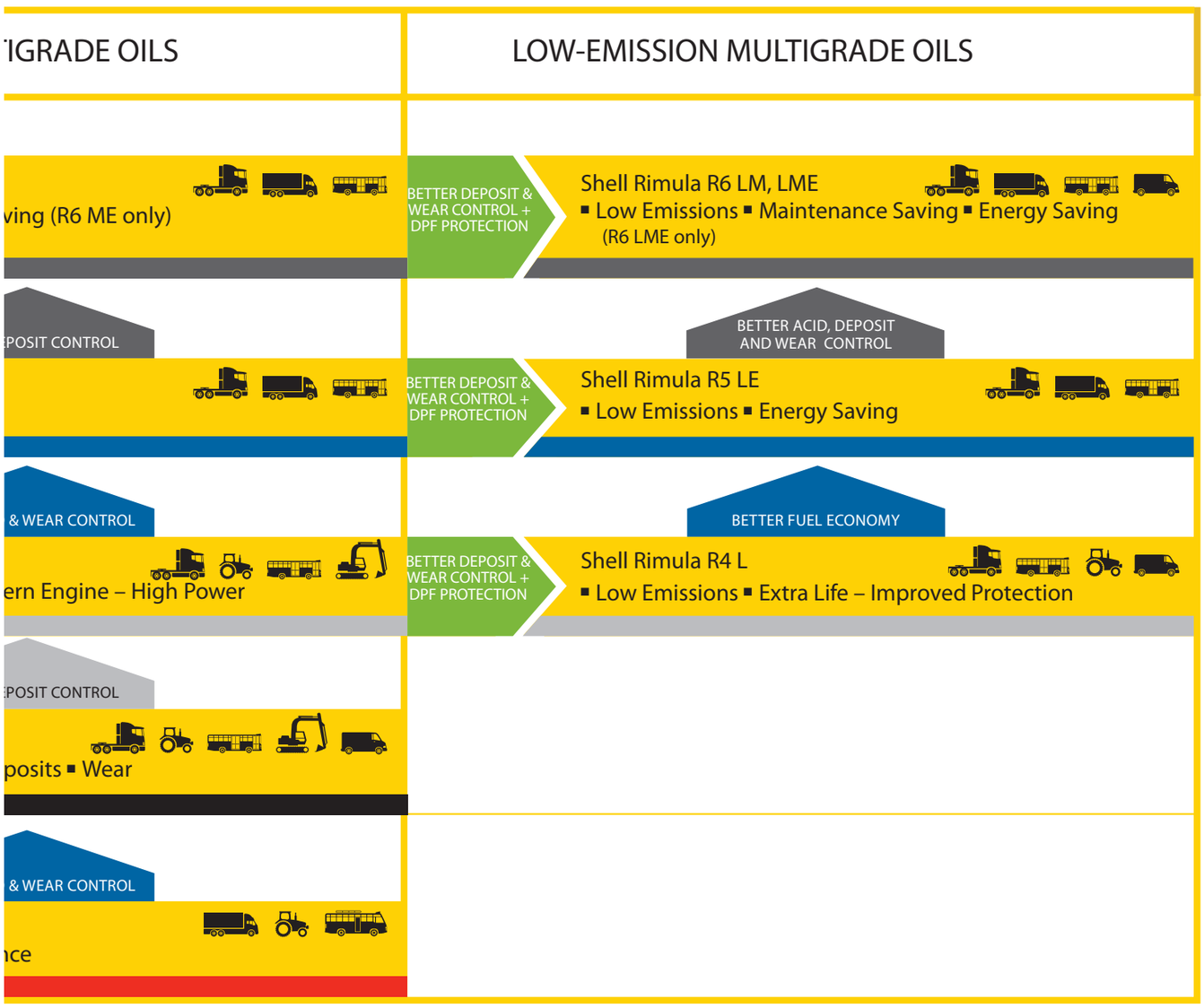
tion oils meet the requirements of on- and off-highway equipment, including the latest



ET IT.

R, AGRICULTURE AND CONSTRUCTION
xtensive experience in supplying heavy duty
oils to some of the world's leading tractor,
ure and construction equipment makers means you
sure we'll have the right product to provide year-
rotection and reliability.

vide range of approvals, products such as Shell Rimula
the ideal choice for heavy duty protection for all your
off-highway equipment. And with approvals from
OEMs such as Caterpillar, Cummins, Deutz and MTU,
products – such as Shell Rimula R4 L – are ready for
t generation of off-highway engines.

Grades with suffices L, M, or E
deliver the following value-added performance:

L = LOW EMISSIONS
Enhanced compatibility with latest-technology engines

M = MAINTENANCE SAVING
Meets long-drain requirements for many engine makers

E = ENERGY SAVING
Demonstrated fuel-saving capability compared to
standard SAE 15W-40 oils

RANGE OF BUSINESS SERVICES AND MAINTENANCE SOLUTIONS

Through a thorough assessment of lubrication-related maintenance practices, Shell works with you to provide a customised recommendation that delivers efficiencies in your equipment maintenance and operation.

Shell LubeAnalyst

Shell LubeAnalyst tests the condition of your lubricant to monitor and track your equipment performance.

Shell LubeAdvisor

Shell LubeAdvisor delivers technical advice, basic product recommendations, and troubleshooting support for lubricant-related equipment problems.

Shell LubePlanner

Shell LubePlanner offers lubrication planning software.

Shell LubeCoach

Shell LubeCoach provides relevant training on topics such as storage and handling of lubricants, lubricant cleanliness, lubrication and the application of lubrication.

Shell LubeExpert

Shell LubeExpert provides specialist on-site technical application support, critical application support and inspections and audits.

Shell LubeMatch

Shell LubeMatch delivers instant professional information online on product recommendation for optimised lubrication of virtually all equipment.

Shell LubeControl

Shell LubeControl offers stock management and Vendor Managed Inventory services.

Shell LubeEquipment

Shell LubeEquipment supports acquisition and set-up of lubricants storage and dispensing equipment.

Shell LubeReclaim

Shell LubeReclaim supports responsible management of used products.

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SIGNIFICANT SAVINGS IN VARIOUS SECTORS THROUGH THE USE OF SHELL RIMULA PRODUCTS

Fleet Sector: Shell Rimula R6 LM and R6 LME

US$188,200 (€130,000) reported saving for fleet company by using fully synthetic oils – Shell Rimula R6 LM and Shell Rimula R6 LME

BVG, Germany's largest bus company, was looking for a lubricant that would help provide increased protection against deposits in order to reduce cleaning intervals of diesel particulate filters (DPF) and reduce maintenance costs. Shell recommended conducting a five-year trial in 15 buses using Shell Rimula R6 LM and Shell Rimula R6 LME. Analysis revealed:

- Reduction in DPF cleaning intervals by 50%
- Extended oil drain intervals while providing excellent wear protection

Construction Sector: Shell Rimula R4

Shell Rimula R4 leads to an estimated US$195,000 in savings per year, through a 40% increase in oil drain intervals

Conirsa is a joint venture company contracted to build Sector 2 of the Interoceanic Highway, a vast road project that will link Peru and Brazil. Conirsa selected Shell for the supply of lubricants for the whole operation: engines, transmissions, final drives, hydraulic and grease applications. Shell LubeClinic was also installed for on-site oil analysis and equipment monitoring. Based on results gathered during more than a year, the customer concluded that using Shell Rimula R4 helped to extend oil drain intervals by 40%, leading to reported annual savings of US$195,000. The benefits included:

- Increase in oil drain intervals from 250 to 350 hours
- Reduction in oil consumption
- Reduction in maintenance costs
- Reduction in truck downtime
- Greater availability of the fleet

Mining Sector: Shell Rimula R3 MV

Changing to Shell Rimula R3 MV increased oil drain intervals of Rio Tinto's truck fleet on an open cast mine by 100%, delivering a reported US$37,500 saving

Rio Tinto, a leading mining and exploration company, was looking to optimise oil drain intervals in its fleet of haul trucks. Shell assessed the situation and advised the change to Shell Rimula R3 MV. As a result, Rio Tinto was able to increase oil drain intervals by 100% and reduce service costs by a reported US$37,500.
The benefits included:

- Extended oil drain intervals from 250 to 500 hours
- Reduced servicing costs
- Increased equipment availability
- Reduced waste oil storage and disposal costs
- Reduced engine oil volumes purchased
- Decreased storage and handling costs

Rio Tinto's annual savings were attributed to using Shell Rimula R3 MV and reducing labour and service component costs.

The savings indicated are specific to the calculation date and mentioned site. These calculations may vary from site to site depending on application, operating conditions, current products being used, condition of the equipment and maintenance practices.

PRODUCT PERFORMANCE

The Shell Rimula Energised Protection range of oils has products to meet virtually all the requirements and approvals of the world's leading manufacturers of heavy duty engines, including Caterpillar, Cummins, Mercedes-Benz, Volvo and many others.

PRODUCT	VISCOSITY GRADES	KEY BENEFITS	VALUE	SPECIFICATIONS AND APPROVALS
Shell Rimula R6 LME	5W-30	▪ Low Emissions ▪ Maintenance Saving ▪ Energy Saving	Delivers up to 2% fuel economy savings*	ACEA: E6, E7; Cummins: CES 20077; MAN: M3477; MB Approval: 228.51.
Shell Rimula R6 LM	10W-40	▪ Low Emissions ▪ Maintenance Saving	Can deliver oil drain intervals up to 120,000kms†	ACEA: E7, E6, E4-99; API: CI-4, CH-4, CG-4, CF-4, CF; Caterpillar: ECF-1-A; Cummins: CES 20077, 72 ,71; DAF: Meets ACEA E6, E4-99; Deutz: DQC IV-05; Mack: EO-M+; MAN: M3477, M3271-1; MB Approval: 228.51, 226.9; MTU: Category 3.1; Renault Trucks: RD-2; Volvo: CNG, VDS-2, Jaso DH-2 (Meets)
Shell Rimula R6 ME	5W-30	▪ Maintenance Saving ▪ Energy Saving		ACEA: E4; API: CF; DAF: HP-1/2; MAN: M3277; MB Approval: 228.5; MTU: Category 3; Volvo: VDS-2, IVECO Standard TFE (Meets)
Shell Rimula R6 M	10W-40	▪ Maintenance Saving		ACEA: E7, E4; API: CF; Cummins: CES 20072; Deutz: DQC IV-05; MAN: M3277; MB Approval: 228.5; Mack: EO-M+; MTU: Category 3; Renault trucks: RXD; Volvo: VDS-3, Scania: LDF-2. IVECO T3 E4 (Meets)
Shell Rimula R5 LE	10W-30	▪ Low Emissions ▪ Energy Saving	Delivers 1.6% fuel economy savings**	API CJ-4, CI-4 Plus, CI-4, CH-4, CG-4; ACEA E9, E7; Caterpillar ECF-3, ECF-2; Cummins CES 20081; Mack: EO-O premium plus; MAN: M 3575; MTU Cat 2.1; JASO DH-2; MB Approval: 228.31; Volvo: VDS-4, Renault VI RLD-3
Shell Rimula R4 L	15W-40	▪ Low Emissions ▪ Extra Life – Improved Protection		API: CJ-4, CI-4+, CI-4, CH-4, CG-4, CF-4, CF, (SM/SL/SJ/SH); ACEA: E9, E7; Caterpillar: ECF-2, ECF-3; Cummins: CES 20081, 77, 72, 71; Deutz: DQC III-05; DDC: 93K218; MACK: EO-O Premium Plus; JASO DH-2, MAN: M3575; MB Approval: 228.31; MTU: Category 2; Renault Trucks: RLD-3; Scania:LDF-LA, Volvo: VDS-4, VDS-3, IVECO T2 E7 (Meets)
Shell Rimula R4	15W-40	▪ Unique Active Technology ▪ Modern Engine – High Power	Lasts over 30% longer***	API: CI-4,CH-4, CG-4, CF-4, CF; ACEA: E7, E5, E3; Global: DHD-1; Caterpillar: ECF-1-A; Cummins: CES 20078, 77, 76, 75, 72, 71; DDC: 93K215; Mack: EO-M, EO-M+; MAN: M3275-1; MB Approval: 228.3; Renault Trucks: RLD-2; Volvo: VDS-3.
Shell Rimula R3 X	15W-40	Triple Action Resists: ▪ Wear ▪ Deposits ▪ Heat	Up to 35% lower wear****	API: CH-4, CG-4, CF-4, CF; ACEA: E5, E3; Caterpillar: ECF-1-A; Cummins: CES 20071, 72, 75, 76, 77; DDC: 93K215; Mack: EO-M, EO-M+; MAN: M3275-1; MB Approval: 228.3; MTU: Category 2; Renault Trucks: RD-2; Volvo: VDS-2.
Shell Rimula R2 Extra	15W-40 20W-50	▪ Turbocharged Performance	Up to 20% more deposit control components *****	API: CF-4, CF; ACEA: E2; MAN: 271; MB Approval: 228.1; Volvo: VDS.
Shell Rimula R2	SAE 30 SAE 40	▪ Tough Long-Lasting Action		API: CF

Specifications and approvals may vary with viscosity grade.
*Compared to a typical SAE 10W-40 oil. **For 10W-30 viscosity grade only, highway cycles, compared to a typical 15W-40.
***Than competitors' products meeting API CH-4 or lower specifications as demonstrated in laboratory oxidation test. ****In on-road heavy duty testing vs. API CG-4 oils.
*****Compared to basic API CF-4 oils. †Depends on engine type and operating conditions. Consult OEM recommendations for oil drain intervals.

GEAR, AXLES & TRANSMISSIONS

ENGINES

BEARINGS

In addition to the full range of Shell Rimula heavy duty engine oils, Shell also offers a complete portfolio of lubricants for every part of your equipment, including Shell Spirax gear, axle and transmission oils and Shell Gadus heavy duty greases.

For more information contact your Shell representative or visit **www.shell.com/lubes**

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